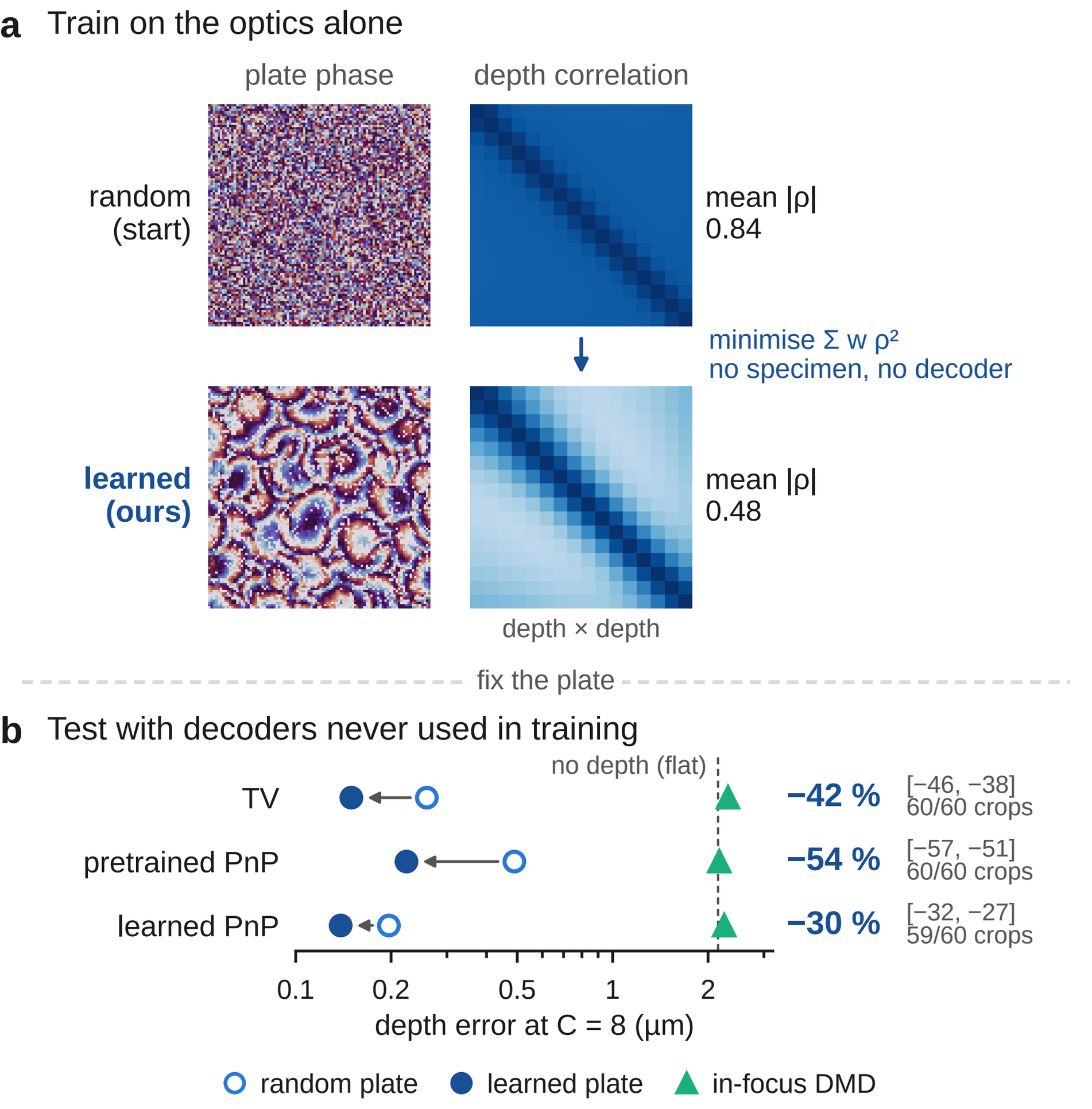

a
Train on the optics alone
plate phase
depth correlation
random (start)
mean |ρ| 0.84
minimise Σ w ρ²
no specimen, no decoder
learned (ours)
mean |ρ| 0.48
depth × depth
fix the plate
b
Test with decoders never used in training
no depth (flat)
TV
pretrained PnP
learned PnP
−42 %
[−46, −38] 60/60 crops
−54 %
[−57, −51] 60/60 crops
−30 %
[−32, −27] 59/60 crops
0.1
0.2
0.5
1
2
depth error at C = 8 (µm)
random plate
learned plate
in-focus DMD